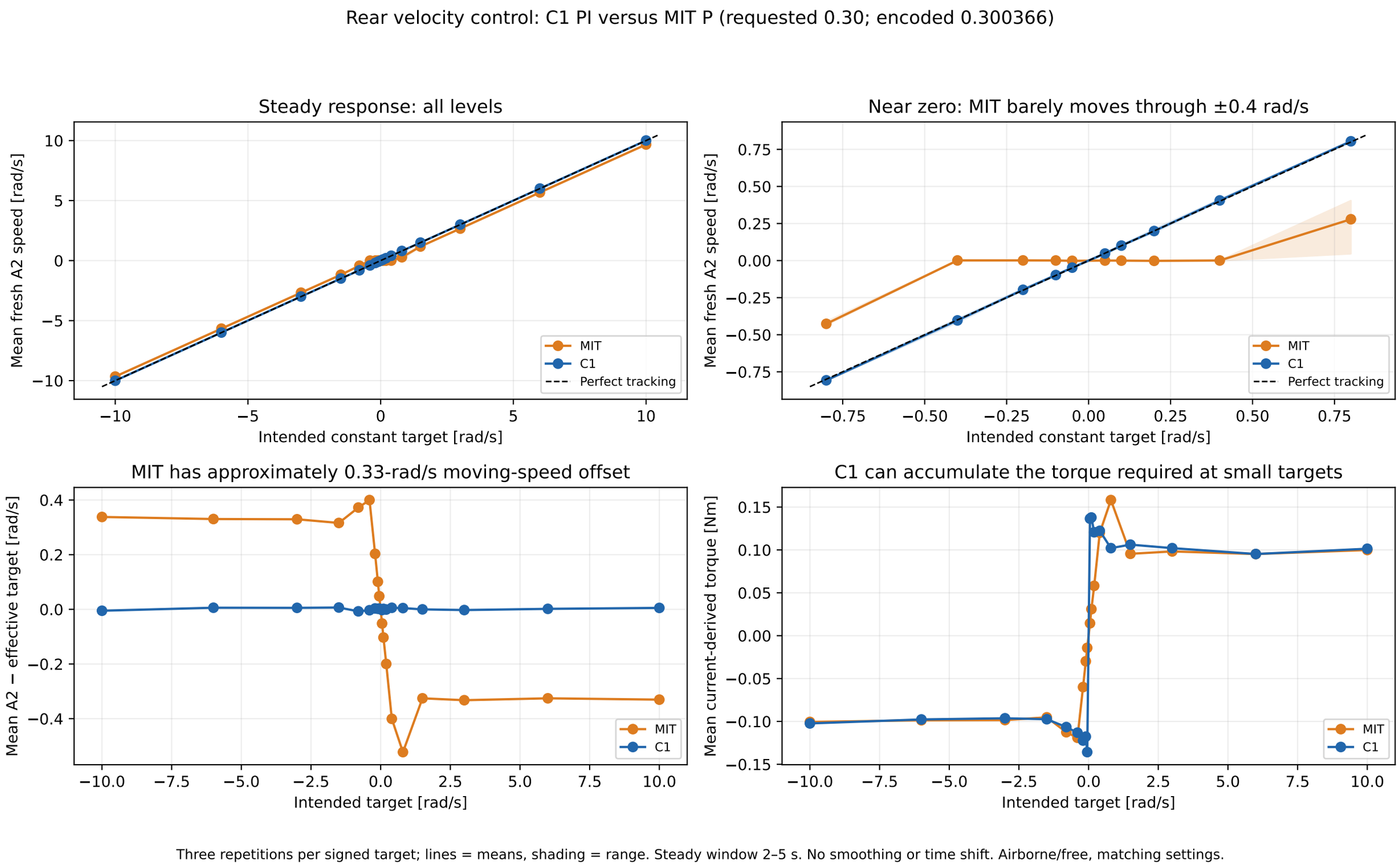
Rear velocity control: C1 PI versus MIT P (requested 0.30; encoded 0.300366)
Steady response: all levels
Mean fresh A2 speed [rad/s]
Intended constant target [rad/s]
MIT
C1
Perfect tracking
Near zero: MIT barely moves through ±0.4 rad/s
Mean fresh A2 speed [rad/s]
Intended constant target [rad/s]
MIT
C1
Perfect tracking
MIT has approximately 0.33-rad/s moving-speed offset
Mean A2 − effective target [rad/s]
Intended target [rad/s]
MIT
C1
C1 can accumulate the torque required at small targets
Mean current-derived torque [Nm]
Intended target [rad/s]
MIT
C1
Three repetitions per signed target; lines = means, shading = range. Steady window 2–5 s. No smoothing or time shift. Airborne/free, matching settings.